



OBJECT ORIENTED SOFTWARE ENGINEERING

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The Process Models

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- Prescriptive process models includes an **orderly approach** to software engineering
 - Organize framework activities in certain order.
- Software engineer choose process framework that includes activities like; **Communication, Planning, Modeling, Construction, Deployment.**

- Each elements are inter-related to one another (called workflow).
- Prescriptive models are also known as “**Traditional models**”.

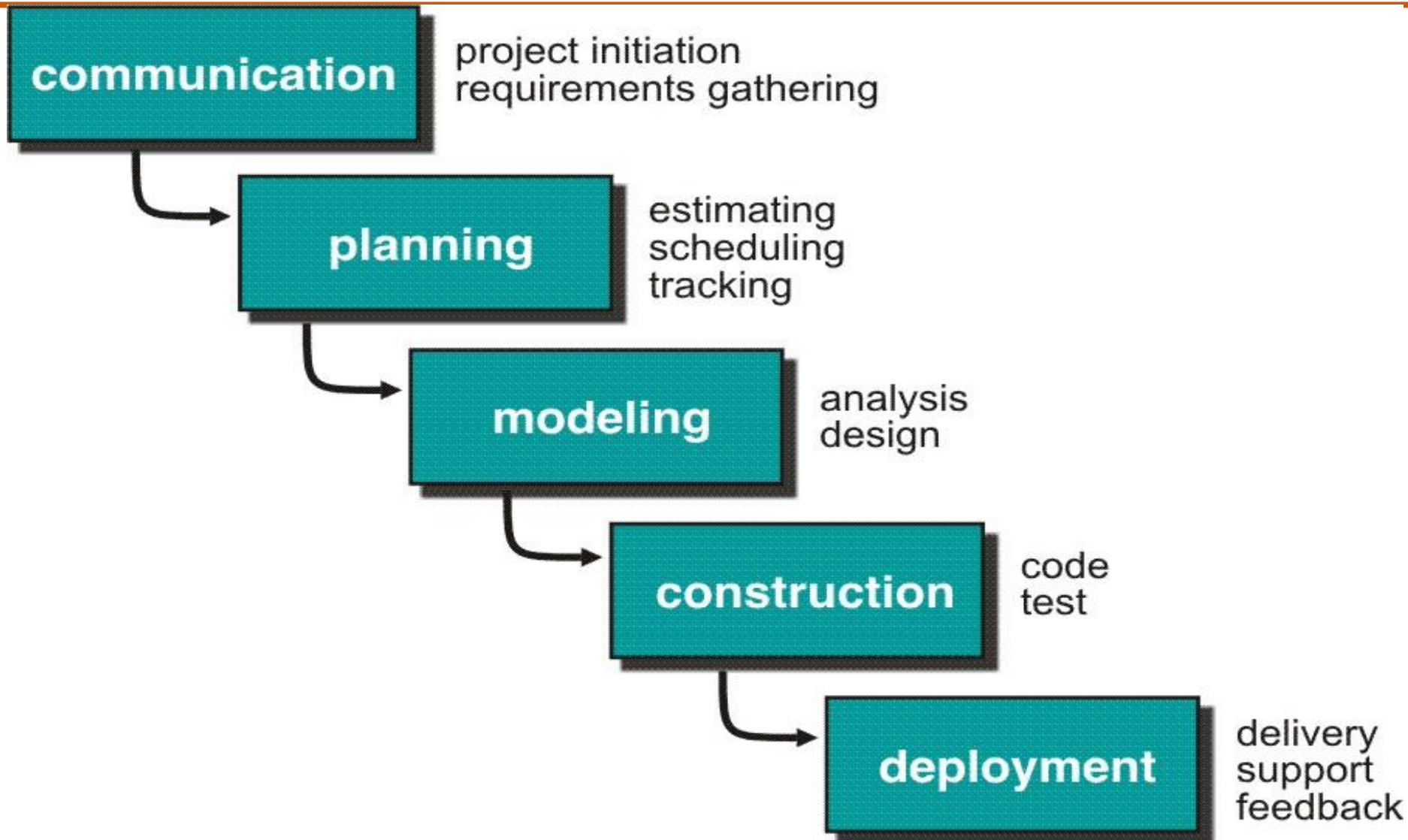
The Prescriptive models are:

- 1. The Waterfall Model.**
- 2. The V-Model.**
- 3. Incremental Process Model.**
- 4. Evolutionary Process Models**
 - Prototyping**
 - Spiral Model**
- 5. The Unified Process Model**

- The **waterfall model**, sometimes called the *classic life cycle*.
- Suggests a systematic **sequential approach** to software development.
- It begins with customer **specification of requirements** and progresses through planning, modeling, construction, deployment.

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Waterfall Model Design



- Waterfall approach was first **SDLC Model** to be used widely in Software Engineering to ensure success of the project.
- In "The Waterfall" approach, the whole process of software development is divided into **separate phases**. In this Waterfall model, typically, the **outcome of one phase acts as the input for the next phase sequentially**.
- The following illustration is a representation of the different phases of the Waterfall Model

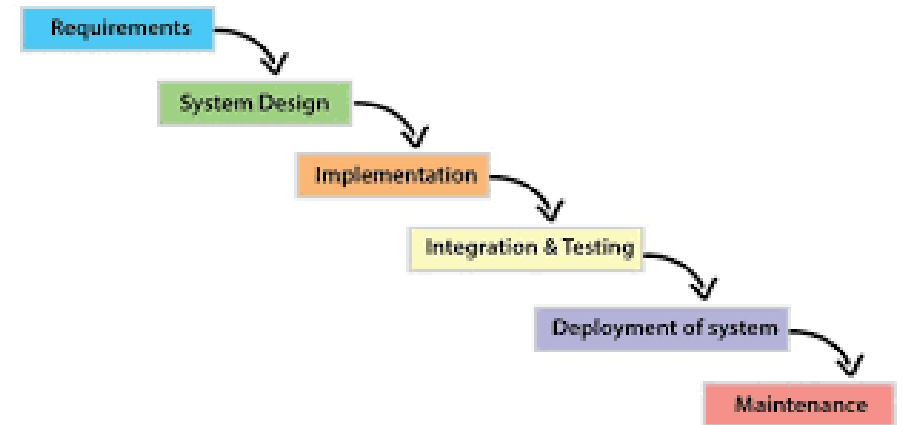
The sequential phases in Waterfall model are –

- **Requirement Gathering and analysis** – All possible requirements of the system to be developed, are captured in this phase and documented in a **requirements specification document**.
- **System Design** – The requirements specification from first phase are studied in this phase and the system design is prepared.
- This system design helps in specifying hardware and system requirements and helps in **defining the overall system architecture**.

- **Implementation** – With inputs from the system design, the system is first developed in small programs called **units**, which are **integrated** in the next phase. Each unit is developed and tested for its functionality, which is referred to as **Unit Testing**.
- **Integration and Testing** – All the units developed in the implementation phase are integrated into a system after testing of each unit. Post integration the entire system is tested for any faults and failures.

- **Deployment of system** – Once the functional and non-functional testing is done; the product is deployed in the customer environment or released into the market.
- **Maintenance** – There are **some issues which come up in the client environment**. To fix those issues, patches are released. Also to enhance the product some better versions are released. Maintenance is done to deliver these changes in the customer environment.

- All these phases are cascaded to each other in which progress is seen as **flowing steadily downwards** (like a waterfall) through these phases.
- The next phase is started only after the defined set of goals are achieved for previous phase and it is signed off, so the name "**Waterfall Model**".
- In this model, **phases do not overlap** and **no backtracking** possible.



- Every software developed is different and requires a suitable SDLC approach to be followed based on the internal and external factors.

Some situations where the use of Waterfall model is most appropriate are –

- Requirements are very well documented, clear and fixed.
- Product definition is stable.
- Technology is understood and is not dynamic.
- There are no ambiguous requirements.
- Ample resources with required expertise are available to support the product.
- The project is short.

Some of the major **advantages** of the Waterfall Model are as follows –

- Simple and easy to understand and also to use.
- Easy to manage. Due to the rigidity of the model each phase has specific deliverables and a review process.
- Phases are processed and completed one at a time.
- Works well for smaller projects where requirements are very well understood.
- Clearly defined stages.
- Well understood milestones.
- Easy to arrange tasks.
- Process and results are well documented.

The major **disadvantages** of the Waterfall Model are as follows –

- No working software is produced until late during the life cycle.
- High amounts of risk and uncertainty.
- Not a good model for complex and object-oriented projects.
- Poor model for long and ongoing projects.
- Not suitable for the projects where requirements are at a moderate to high risk of changing. So, risk and uncertainty is high with this process model.

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- It is difficult to measure progress within stages.
 - Cannot accommodate changing requirements.
 - Adjusting scope during the life cycle can end a project.
 - Integration is done as a "big-bang. at the very end, which doesn't allow identifying any technological or business bottleneck or challenges early.



THANK YOU

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